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Printhead design with multiple fluid paths to jetting channels

Abstract

Printheads and design of printheads. In one embodiment, a printhead comprises a plurality of jetting channels, and a manifold apparatus fluidly coupled to the jetting channels. For each jetting channel, the printhead includes a first fluid path between the jetting channel and the manifold apparatus, and a second fluid path between the jetting channel and the manifold apparatus. The jetting channel is configured to jet a print fluid via pressure waves generated in a pressure chamber of the jetting channel. Lengths of the first fluid path and the second fluid path are different by a threshold length so that an arrival time of the pressure waves at the manifold apparatus are different by a threshold time.

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Background/Summary

RELATED APPLICATIONS (1) This non-provisional patent application is a continuation of U.S. patent application Ser. No. 17/669,350 filed on Feb. 10, 2022, which is incorporated herein by reference.

TECHNICAL FIELD

(1) The following disclosure relates to the field of image formation, and in particular, to printheads and the design of printheads.

BACKGROUND

(2) Image formation is a procedure whereby a digital image is recreated by propelling droplets of ink or another type of print fluid onto a medium, such as paper, plastic, a substrate for 3D printing, etc. Image formation is commonly employed in apparatuses, such as printers (e.g., inkjet printer), facsimile machines, copying machines, plotting machines, multifunction peripherals, etc. The core of a typical jetting apparatus or image forming apparatus is one or more liquid-droplet ejection heads (referred to generally herein as “printheads”) having nozzles that discharge liquid droplets, a mechanism for moving the printhead and/or the medium in relation to one another, and a controller that controls how liquid is discharged from the individual nozzles of the printhead onto the medium in the form of pixels.

(3) A typical printhead includes a plurality of nozzles aligned in one or more rows along a discharge surface of the printhead. Each nozzle is part of a “jetting channel”, which includes the nozzle, a pressure chamber, and a diaphragm that vibrates in response to an actuator, such as a piezoelectric actuator. A printhead also includes a driver circuit that controls when each individual jetting channel fires based on image or print data. To jet from a jetting channel, the driver circuit

provides a jetting pulse to the actuator, which causes the actuator to deform a wall of the pressure chamber (i.e., the diaphragm). The deformation of the pressure chamber creates pressure waves within the pressure chamber that eject a droplet of print fluid (e.g., ink) out of the nozzle.

(4) Multiple jetting channels within a printhead are fluidly coupled to a common fluid path that conveys the print fluid, which is referred to as a manifold. One problem encountered within printheads is that pressure waves may escape from the jetting channels, and propagate along the manifold. The pressure waves in the manifold can affect jetting in individual jetting channels, which can result in jetting instability.

SUMMARY

(5) Embodiments described herein provide for printheads and the design of printheads having multiple fluid paths between a manifold apparatus and jetting channels. The pressure waves that escape from the jetting channels propagate back towards the manifold apparatus along the different fluid paths. The fluid paths are designed so that there is a difference between the lengths of the fluid paths by a threshold length so that the arrival time of the pressure waves at the manifold apparatus is different by a threshold time. One advantage is that the pressure waves arriving at different times can at least partially cancel each other out within the manifold apparatus. This can result in improved jetting consistency and performance.

(6) One embodiment comprises a printhead comprising a plurality of jetting channels, and a manifold apparatus fluidly coupled to the jetting channels. For each jetting channel of the plurality, the printhead includes a first fluid path between the jetting channel and the manifold apparatus, and a second fluid path between the jetting channel and the manifold apparatus. The jetting channel is configured to jet a print fluid via pressure waves generated in a pressure chamber of the jetting channel. Lengths of the first fluid path and the second fluid path are different by a threshold length so that an arrival time of the pressure waves at the manifold apparatus are different by a threshold time.

(7) One embodiment comprises a method of operating a printhead comprising a plurality of jetting channels configured to jet a print fluid. For each jetting channel of the plurality, the method comprises conveying the print fluid between a manifold apparatus and the jetting channel over a first fluid path, conveying the print fluid between the manifold apparatus and the jetting channel over a second fluid path, generating pressure waves in a pressure chamber of the jetting channel that propagate along the first fluid path and the second fluid path, and producing a difference in arrival time of the pressure waves at the manifold apparatus by a threshold time due to a difference in length between the first fluid path and the second fluid path by a threshold length.

(8) One embodiment comprises a design tool for a printhead comprising a plurality of jetting channels configured to jet a print fluid, and a manifold apparatus fluidly coupled to the jetting channels. The design tool comprises at least one processor and memory, and the processor causes the design tool to design a first fluid path between the manifold apparatus and a jetting channel having a pressure chamber configured to jet based on pressure waves, design a second fluid path between the manifold apparatus and the jetting channel, and select a target difference in arrival time of the pressure waves that propagate along the first fluid path and arrive at the manifold apparatus, and the pressure waves that propagate along the second fluid path and arrive at the manifold apparatus. The processor further causes the design tool to select a difference in length between the first fluid path and the second fluid path by a threshold length that causes the target difference in arrival time of the pressure waves at the manifold apparatus, and configure the first fluid path and the second fluid path for the jetting channels based on the threshold length.

(9) One embodiment comprises a method of operating a printhead in non-circulation mode, where the printhead comprises a plurality of jetting channels configured to jet a print fluid. For each jetting channel of the plurality, the method comprises conveying the print fluid from a manifold to the jetting channel over a first fluid path, conveying the print fluid from the manifold to the jetting channel over a second fluid path, generating pressure waves in a pressure chamber of the jetting

channel that propagate along the first fluid path and the second fluid path, and producing a difference in arrival time of the pressure waves at the manifold by a threshold time due to a difference in length between the first fluid path and the second fluid path by a threshold length.

(10) One embodiment comprises a method of operating a printhead in circulation mode, where the printhead comprises a plurality of jetting channels configured to jet a print fluid. For each jetting channel of the plurality, the method comprises conveying the print fluid from a first manifold to the jetting channel over a first fluid path, conveying non-jetted print fluid from the jetting channel to a second manifold over a second fluid path, generating pressure waves in a pressure chamber of the jetting channel that propagate along the first fluid path and the second fluid path, and producing a difference in arrival time of the pressure waves at the first manifold and at the second manifold by a threshold time due to a difference in length between the first fluid path and the second fluid path by a threshold length.

(11) The above summary provides a basic understanding of some aspects of the specification. This summary is not an extensive overview of the specification. It is intended to neither identify key or critical elements of the specification nor delineate any scope particular embodiments of the specification, or any scope of the claims. Its sole purpose is to present some concepts of the specification in a simplified form as a prelude to the more detailed description that is presented later.

Description

DESCRIPTION OF THE DRAWINGS

- (1) Some embodiments of the present disclosure are now described, by way of example only, and with reference to the accompanying drawings. The same reference number represents the same element or the same type of element on all drawings.
- (2) FIG. 1 is a schematic diagram of a jetting apparatus in an illustrative embodiment.
- (3) FIG. 2 is a perspective view of a printhead in an illustrative embodiment.
- (4) FIG. 3 is a perspective view of a printhead in an illustrative embodiment.
- (5) FIG. 4 is a cross-sectional view of a printhead in an illustrative embodiment.
- (6) FIG. 5 is another cross-sectional view of a portion of a printhead in an illustrative embodiment.
- (7) FIG. 6 is a schematic diagram of a printhead in an illustrative embodiment.
- (8) FIG. 7 is a schematic diagram of a manifold apparatus and a jetting channel in an illustrative embodiment.
- (9) FIG. 8 is a flow chart illustrating a method of operating a printhead in an illustrative embodiment.
- (10) FIG. 9 is a cross-sectional view of a portion of a printhead in an illustrative embodiment.
- (11) FIG. 10 is a schematic diagram of a printhead in an illustrative embodiment.
- (12) FIG. 11 is a schematic diagram of a manifold and a jetting channel in an illustrative embodiment.
- (13) FIG. 12 is a flow chart illustrating a method of operating a printhead in non-circulation mode in an illustrative embodiment.
- (14) FIG. 13 illustrates an exploded, perspective view of a head member of a printhead in an illustrative embodiment.
- (15) FIGS. 14-15 are cross-sectional views of a portion of a printhead in illustrative embodiments.
- (16) FIG. 16 is a schematic diagram of a printhead in an illustrative embodiment.
- (17) FIG. 17 is a schematic diagram of manifolds and a jetting channel in an illustrative embodiment.
- (18) FIG. 18 is a flow chart illustrating a method of operating a printhead in circulation mode in an illustrative embodiment.

- (19) FIGS. **19-27** are schematic diagrams of a printhead in illustrative embodiments.
- (20) FIG. **28** is a cross-sectional view of a portion of a printhead in an illustrative embodiment.
- (21) FIG. **29** is a schematic diagram of a printhead in an illustrative embodiment.
- (22) FIG. **30** illustrates an exploded, perspective view of a head member of a printhead in an illustrative embodiment.
- (23) FIG. **31** is a cross-sectional view of a portion of a printhead in an illustrative embodiment.
- (24) FIG. **32** illustrates an exploded, perspective view of a head member of a printhead in an illustrative embodiment.
- (25) FIG. **33** is a cross-sectional view of a portion of a printhead in an illustrative embodiment.
- (26) FIG. **34** is a cross-sectional view of a portion of a printhead in an illustrative embodiment.
- (27) FIG. **35** is a schematic diagram of a design tool for a printhead in an illustrative embodiment.
- (28) FIG. **36** is a flow chart illustrating a method of designing a printhead in an illustrative embodiment.

(29) FIG. **37** illustrates a processing system operable to execute a computer readable medium embodying programmed instructions to perform desired functions in an illustrative embodiment.

DETAILED DESCRIPTION

(30) The figures and the following description illustrate specific exemplary embodiments. It will thus be appreciated that those skilled in the art will be able to devise various arrangements that, although not explicitly described or shown herein, embody the principles of the embodiments and are included within the scope of the embodiments. Furthermore, any examples described herein are intended to aid in understanding the principles of the embodiments, and are to be construed as being without limitation to such specifically recited examples and conditions. As a result, the inventive concept(s) is not limited to the specific embodiments or examples described below, but by the claims and their equivalents.

(31) FIG. **1** is a schematic diagram of a jetting apparatus **100** in an illustrative embodiment. A jetting apparatus **100** is a device or system that uses one or more printheads to eject a print fluid or marking material onto a medium. One example of jetting apparatus **100** is an inkjet printer (e.g., a cut-sheet or continuous-feed printer) that performs single-pass printing. Other examples of jetting apparatus **100** include a scan pass inkjet printer (e.g., a wide format printer), a multifunction printer, a desktop printer, an industrial printer, a 3D printer, etc. Generally, jetting apparatus **100** includes a mount mechanism **102** that supports one or more printheads **104** in relation to a medium **112**. Mount mechanism **102** may be fixed within jetting apparatus **100** for single-pass printing. Alternatively, mount mechanism **102** may be disposed on a carriage assembly that reciprocates back and forth along a scan line or sub-scan direction for multi-pass printing. Printheads **104** are a device, apparatus, or component configured to eject droplets **106** of a print fluid, such as ink (e.g., water, solvent, oil, or UV-curable), through a plurality of nozzles (not visible in FIG. **1**). The droplets **106** ejected from the nozzles of printheads **104** are directed toward medium **112**. Medium **112** comprises any type of material upon which ink or another marking material is applied by a printhead, such as paper, plastic, card stock, transparent sheets, a substrate for 3D printing, cloth, etc. Typically, nozzles of printheads **104** are arranged in one or more rows so that ejection of a print fluid from the nozzles causes formation of characters, symbols, images, layers of an object, etc., on medium **112** as printhead **104** and/or medium **112** are moved relative to one another. Jetting apparatus **100** may include a media transport mechanism **114** or a media holding bed **116**. Media transport mechanism **114** is configured to move medium **112** relative to printheads **104**. Media holding bed **116** is configured to support medium **112** in a stationary position while the printheads **104** move in relation to medium **112**.

(32) Jetting apparatus **100** also includes a jetting apparatus controller **122** that controls the overall operation of jetting apparatus **100**. Jetting apparatus controller **122** may connect to a data source to receive print data, image data, or the like, and control each printhead **104** to discharge the print fluid on medium **112**. Jetting apparatus **100** also includes one or more reservoirs **124** for a print

fluid or multiple types of print fluid. Although not shown in FIG. 1, reservoirs **124** are fluidly coupled to printheads **104**, such as with hoses or the like.

(33) FIG. 2 is a perspective view of a printhead **104** in an illustrative embodiment. In this embodiment, printhead **104** includes a head member **202** and electronics **204**. Head member **202** is an elongated component that forms the jetting channels of printhead **104**. A typical jetting channel includes a nozzle, a pressure chamber, and a diaphragm that is driven by an actuator, such as a piezoelectric actuator. Electronics **204** control how the nozzles of printhead **104** jet droplets in response to data signals and control signals received from another controller (e.g., jetting apparatus controller **122**). Electronics **204** include an embedded printhead controller **206** or driver circuits configured to drive individual jetting channels based on the data signals and control signals. The bottom surface of head member **202** in FIG. 2 includes the nozzles of the jetting channels, and represents the discharge surface **220** of printhead **104**. The top surface of head member **202** in FIG. 2 (referred to as I/O surface **222**) represents the Input/Output (I/O) portion for receiving one or more print fluids into printhead **104**, and/or conveying print fluids (e.g., fluids that are not jetted) out of printhead **104**. I/O surface **222** includes a plurality of I/O ports **211-214**. An I/O port **211-214** may comprise an inlet I/O port, which is an opening in head member **202** that acts as an entry point for a print fluid. An I/O port **211-214** may comprise an outlet I/O port, which is an opening in head member **202** that acts as an exit point for a print fluid. I/O ports **211-214** may include a hose coupling, hose barb, etc., for coupling with a hose of a reservoir, a cartridge, or the like. The number of I/O ports **211-214** is provided as an example, as printhead **104** may include other numbers of I/O ports.

(34) Head member **202** includes a housing **230** and a plate stack **232**. Housing **230** is a rigid member made from stainless steel or another type of material. Housing **230** includes an access hole **234** that provides a passageway for electronics **204** to pass through housing **230** so that actuators may interface with (i.e., come into contact with) diaphragms of the jetting channels. Plate stack **232** attaches to an interface surface (not visible) of housing **230**. Plate stack **232** (also referred to as a laminate plate stack) is a series of plates that are fixed or bonded to one another to form a laminated stack. Plate stack **232** may include the following plates: one or more nozzle plates, one or more chamber plates, restrictor plates, and a diaphragm plate. A nozzle plate includes a plurality of nozzles that are arranged in one or more rows (e.g., two rows, four rows, etc.). A chamber plate includes a plurality of openings that form the pressure chambers of the jetting channels. A restrictor plate includes a plurality of restrictors that fluidly connect the pressure chambers of the jetting channels with a manifold. A diaphragm plate is a sheet of a semi-flexible material that vibrates in response to actuation by an actuator (e.g., piezoelectric actuator). The embodiment in FIG. 2 illustrates one particular configuration of a printhead **104**, and it is understood that other printhead configurations are considered herein that have a plurality of jetting channels.

(35) FIG. 3 is a perspective view of printhead **104** in an illustrative embodiment. In FIG. 3, plate stack **232** is attached or affixed to housing **230**. FIG. 4 is a cross-sectional view of printhead **104** in an illustrative embodiment. FIG. 4 shows a cross-section of a portion of row of jetting channels **402** along cut-plane 4-4 in FIG. 3. A jetting channel **402** is a structural element within printhead **104** that jets or ejects a print fluid. Each jetting channel **402** includes a diaphragm **410**, a pressure chamber **412**, and a nozzle **414**. An actuator **416** contacts diaphragm **410** to control jetting from a jetting channel **402**. Jetting channels **402** may be formed in one or more rows along a length of printhead **104**, and each jetting channel **402** may have a similar configuration as shown in FIG. 4.

(36) FIG. 5 is another cross-sectional view of a portion of printhead **104** in an illustrative embodiment. FIG. 5 shows a cross-section of printhead **104** along cut-plane 5-5 in FIG. 3. As in FIG. 4, jetting channel **402** includes diaphragm **410**, pressure chamber **412**, and nozzle **414**. A manifold apparatus **518** (also referred to as a manifold assembly) of printhead **104** is fluidly coupled to jetting channel **402** to supply a print fluid to jetting channel **402** (and other jetting channels **402** of printhead **104** configured to jet the same type of print fluid), and/or to receive non-

jetted print fluid from jetting channel **402**. Pressure chamber **412** is fluidly coupled to manifold apparatus **518** through a restrictor **520** (which may also be referred to as a first restrictor, a top restrictor, etc.). Restrictor **520** controls a flow of print fluid between manifold apparatus **518** and pressure chamber **412** along one fluid path. In this embodiment, pressure chamber **412** is also fluidly coupled to manifold apparatus **518** through another restrictor **522**. Restrictor **522** controls a flow of print fluid between manifold apparatus **518** and pressure chamber **412** along another fluid path. One wall of pressure chamber **412** is formed with diaphragm **410** that physically interfaces with actuator **416**. Diaphragm **410** may comprise a sheet of semi-flexible material that vibrates in response to actuation by actuator **416**. The print fluid flows through pressure chamber **412** and out of nozzle **414** in the form of a droplet in response to actuation by actuator **416**. Actuator **416** is configured to receive a jetting pulse, and to actuate or “fire” in response to the jetting pulse. Firing of actuator **416** in jetting channel **402** creates pressure waves in pressure chamber **412** that cause jetting of a droplet from nozzle **414**.

(37) A jetting channel **402** as shown in FIGS. 4-5 are examples to illustrate a basic structure of a jetting channel, such as the diaphragm, pressure chamber, and nozzle. Other types of jetting channels are also considered herein. For example, some jetting channels may have a pressure chamber having a different shape than is illustrated in FIGS. 4-5. Also, the position of a manifold apparatus **518**, restrictors **520/522**, diaphragm **410**, etc., may differ in other embodiments.

(38) FIG. 6 is a schematic diagram of a printhead **104** in an illustrative embodiment. A plurality of jetting channels **402** of printhead **104** is schematically illustrated in FIG. 6 as a row of nozzles **414** fluidly coupled to manifold apparatus **518**. As will be described in more detail below, a manifold apparatus **518** may comprise one or more manifolds. A manifold is a conduit or channel internal to printhead **104** (i.e., within the main body or housing **230** of printhead **104**) that provides a common fluid pathway for a plurality of jetting channels **402**. For each of the jetting channels **402** illustrated, there is a first fluid path **601** (also referred to as fluid conduit, fluid channel, etc.) between the jetting channel **402** and manifold apparatus **518**, and a second fluid path **602** between the jetting channel **402** and the manifold apparatus **518**. In the embodiment shown in FIG. 5, for example, the first fluid path **601** between jetting channel **402** and manifold apparatus **518** may be through restrictor **520**, which controls the flow of print fluid along the first fluid path **601**. Further, the second fluid path **602** between jetting channel **402** and manifold apparatus **518** may be through restrictor **522**, which controls the flow of print fluid along the second fluid path **602**. Thus, the first fluid path **601** and the second fluid path **602** represent distinct pathways for the print fluid to flow between pressure chamber **412** and manifold apparatus **518**.

(39) FIG. 7 is a schematic diagram of manifold apparatus **518** and a jetting channel **402** in an illustrative embodiment. FIG. 7 shows the first fluid path **601** between jetting channel **402** and manifold apparatus **518**, and the second fluid path **602** between jetting channel **402** and manifold apparatus **518**. The first fluid path **601** has a length **701**, and the second fluid path **602** has a length **702**. In this embodiment, the length **701** of the first fluid path **601** is different than the length **702** of the second fluid path **602** by a threshold length (e.g., millimeters). When actuator **416** fires in response to a jetting pulse, pressure waves **706** are created in pressure chamber **412** that cause jetting of a droplet from nozzle **414**. These pressure waves **706** may escape pressure chamber **412** and propagate along the first fluid path **601** and the second fluid path **602** toward manifold apparatus **518**. The pressure waves **706** are initially in-phase when escaping the pressure chamber **412**. The length **701** of the first fluid path **601** is different than the length **702** of the second fluid path **602** by the threshold length so that the arrival time of pressure waves **706** at manifold apparatus **518** are not equal and are different by a threshold time (e.g., milliseconds). Thus, the pressure waves **706** propagated along the first fluid path **601** are out-of-phase with the pressure waves **706** propagated along the second fluid path **602** when received at manifold apparatus **518**. One technical benefit is when the pressure waves **706** pass through each other or interfere within manifold apparatus **518**, the pressure waves **706** interfere destructively. As described in the

background, the pressure waves **706** that escape from the jetting channels **402** can propagate along manifold apparatus **518**, which can affect jetting in individual jetting channels **402**. If the pressure waves **706** escaping along the first fluid path **601** and the second fluid path **602** were in-phase when received at manifold apparatus **518**, then constructive interference would occur within manifold apparatus **518** and the resultant wave would have an amplitude comprising the sum of the maxima of the pressure waves **706** traveling along the first fluid path **601** and the second fluid path **602**. However, when the arrival time of pressure waves **706** at manifold apparatus **518** are different by the threshold time, the pressure waves **706** interfere destructively and the resultant wave has a reduced amplitude.

(40) In one embodiment, the length **701** of the first fluid path **601** is from an origin **711** of the pressure waves **706** to an opening **731** of manifold apparatus **518**. The origin **711** of the pressure waves **706** may be the center **722** of actuator **416**, the center **722** of diaphragm **410** within jetting channel **402**, etc. Similarly, the length **702** of the second fluid path **602** is from the origin **711** of the pressure waves **706** to an opening **732** of manifold apparatus **518**. In one embodiment, the threshold length and/or threshold time may be based on a resonant frequency of the jetting channel **402**. When actuator **416** displaces in response to a jetting pulse, the pressure waves **706** will resonate or absorb at a characteristic frequency. This characteristic frequency is determined by the geometry of pressure chamber **412** (and other structures of a jetting channel **402**) and their associated fluidic properties, and is referred to as the resonant frequency or Helmholtz frequency of a jetting channel **402**. The difference in length of the first fluid path **601** and the second fluid path **602** by the threshold length causes a difference in arrival time of the pressure waves **706** at manifold apparatus **518** by the threshold time. In one embodiment, the threshold time and/or threshold length is based on the resonant frequency or Helmholtz frequency of the jetting channels **402**. For example, the threshold time may be a half resonant cycle (e.g., 0.5) or half Helmholtz cycle of the jetting channels **402**, or a multiple of the half resonant cycle (e.g., 1.5, 2.5, 3.5, etc.). When the threshold time is a half resonant cycle or a multiple of the half resonant cycle, the pressure waves **706** escaping along the first fluid path **601** and the second fluid path **602** would be approximately 180° out-of-phase when they interfere within manifold apparatus **518**. Thus, destructive interference would occur within manifold apparatus **518** and the resultant wave would have little or no amplitude. However, a phase difference other than 180° out-of-phase still results in the pressure waves **706** interfering destructively so that the resultant wave has a reduced amplitude.

(41) In one embodiment, other differences in the features of the first fluid path **601** and the second fluid path **602** may affect the arrival time of pressure waves **706** at manifold apparatus **518**, which are considered herein. For example, material properties of printhead **104** that form the first fluid path **601** and the second fluid path **602** may be different. The volume of the first fluid path **601** and the second fluid path **602** along their respective lengths may be different. Steps or variations along the lengths of the first fluid path **601** and the second fluid path **602** may be different. One or more combinations of these and other features may further affect the arrival time of pressure waves **706** at manifold apparatus **518**.

(42) FIG. **8** is a flow chart illustrating a method **800** of operating printhead **104** in an illustrative embodiment. The steps of method **800** will be described with reference to printhead **104** in FIGS. **4-7**, but those skilled in the art will appreciate that method **800** may be performed by other printheads. Also, the steps of the flow charts described herein are not all inclusive and may include other steps not shown, and the steps may be performed in an alternative order.

(43) For method **800**, it is assumed that printhead **104** includes a plurality of jetting channels **402** fluidly coupled to a manifold apparatus **518**. For each jetting channel **402**, a print fluid is conveyed between manifold apparatus **518** and the jetting channel **402** over a first fluid path **601** (step **802**), and the print fluid is conveyed between manifold apparatus **518** and the jetting channel **402** over a second fluid path **602** (step **804**). Pressure waves **706** are generated in a pressure chamber **412** of the jetting channel **402** (step **806**), such as due to actuation of an actuator **416**, to jet droplets of the

print fluid from a nozzle **414** of the jetting channel **402**. The pressure waves **706** generated in the pressure chamber **412** propagate along the first fluid path **601** to manifold apparatus **518**, and propagate along the second fluid path **602** to manifold apparatus **518**. The design of printhead **104** produces, creates, or generates a difference in arrival time of pressure waves **706** at manifold apparatus **518** (i.e., by a threshold time) due to the difference in length **701** of the first fluid path **601** and length **702** of the second fluid path **602** by the threshold length (step **808**). Thus, the pressure waves **706** that arrive at manifold apparatus **518** over the first fluid path **601** and over the second fluid path **602** interfere destructively within manifold apparatus **518**.

(44) FIGS. **9-12** disclose a printhead **104** in non-circulation mode in one embodiment. In non-circulation mode, print fluid is supplied to a jetting channel **402** through the first fluid path **601** and the second fluid path **602**. FIG. **9** is a cross-sectional view of a portion of printhead **104** in an illustrative embodiment. FIG. **9** shows a cross-section of printhead **104** along cut-plane **5-5** in FIG. **3**. In this embodiment, manifold apparatus **518** comprises a manifold **910** that acts as a common fluid supply for a plurality of jetting channels **402**. Pressure chamber **412** is fluidly coupled to manifold **910** through restrictor **520**, which controls a flow of print fluid from manifold **910** to pressure chamber **412** along one fluid path. Pressure chamber **412** is also fluidly coupled to manifold **910** through restrictor **522**, which controls a flow of print fluid from manifold **910** to pressure chamber **412** along another fluid path.

(45) The arrows in FIG. **9** illustrate a flow of a print fluid from manifold **910** to jetting channel **402**. The print fluid flows from manifold **910** and into pressure chamber **412** through restrictor **520**, and also flows from manifold **910** and into pressure chamber **412** through restrictor **522**. One wall of pressure chamber **412** is formed with diaphragm **410** that physically interfaces with actuator **416**, and vibrates in response to actuation by actuator **416**. The print fluid in pressure chamber **412** is jetted out of nozzle **414** in the form of a droplet in response to actuation by actuator **416**.

(46) FIG. **10** is a schematic diagram of a printhead **104** in an illustrative embodiment. A plurality of jetting channels **402** of printhead **104** is schematically illustrated in FIG. **10** as a row of nozzles **414**. Manifold **910** is a conduit or channel internal to printhead **104** that conveys a print fluid to the jetting channels **402**. Manifold **910** is disposed between I/O ports **211-212** that define inlets of print fluid into printhead **104**. Thus, when print fluid enters printhead **104** at one or both of I/O ports **211-212**, the print fluid flows through manifold **910** to jetting channels **402**. A manifold **910** that conveys a print fluid to jetting channels **402** may be considered as having a direct fluid coupling with the jetting channels **402**, as the manifold **910** is fluidly coupled through a restrictor or similar element that controls the flow of print fluid from manifold **910** to a jetting channel **402**. The major portion or section of manifold **910** is disposed longitudinally within printhead **104** to fluidly couple with the jetting channels **402**. Although one manifold **910** is illustrated in FIG. **10**, a printhead **104** may include more manifolds as desired.

(47) For each of the jetting channels **402** illustrated, there is a first fluid path **601** between the jetting channel **402** and manifold **910**, and a second fluid path **602** between the jetting channel **402** and manifold **910**. In the embodiment shown in FIG. **9**, for example, the first fluid path **601** between jetting channel **402** and manifold **910** may be through restrictor **520**, which controls the flow of print fluid along the first fluid path **601**. Further, the second fluid path **602** between jetting channel **402** and manifold **910** may be through restrictor **522**, which controls the flow of print fluid along the second fluid path **602**.

(48) FIG. **11** is a schematic diagram of manifold **910** and a jetting channel **402** in an illustrative embodiment. FIG. **11** shows the first fluid path **601** between jetting channel **402** and manifold **910**, and the second fluid path **602** between jetting channel **402** and manifold **910**. In one embodiment, the length **701** of the first fluid path **601** is from an origin **711** of the pressure waves **706** to an opening **1131** of manifold **910**. The length **702** of the second fluid path **602** is from the origin **711** of the pressure waves **706** to an opening **1132** of manifold **910**. As above, the length **701** of the first fluid path **601** is different than the length **702** of the second fluid path **602** by a threshold length.

The length **701** of the first fluid path **601** is different than the length **702** of the second fluid path **602** by the threshold length so that the arrival time of pressure waves **706** at manifold **910** are not equal and are different by a threshold time. Thus, when the pressure waves **706** pass through each other or interfere within manifold **910**, the pressure waves **706** interfere destructively.

(49) FIG. **12** is a flow chart illustrating a method **1200** of operating printhead **104** in non-circulation mode in an illustrative embodiment. For method **1200**, it is assumed that printhead **104** includes a plurality of jetting channels **402** fluidly coupled to a manifold **910**. For each jetting channel **402**, a print fluid is conveyed from manifold **910** to the jetting channel **402** over a first fluid path **601** (step **1202**), and the print fluid is conveyed from manifold **910** to the jetting channel **402** over a second fluid path **602** (step **1204**). Pressure waves **706** are generated in a pressure chamber **412** of the jetting channel **402** (step **1206**), such as due to actuation of an actuator **416**, to jet droplets of the print fluid from a nozzle **414** of the jetting channel **402**. The pressure waves **706** generated in the pressure chamber **412** propagate along the first fluid path **601** to manifold **910**, and propagate along the second fluid path **602** to manifold **910**. The design of printhead **104** produces, creates, or generates a difference in arrival time of pressure waves **706** at manifold **910** (i.e., by a threshold time) due to the difference in length **701** of the first fluid path **601** and length **702** of the second fluid path **602** by the threshold length (step **1208**). Thus, the pressure waves **706** that arrive at manifold **910** over the first fluid path **601** and over the second fluid path **602** interfere destructively within manifold **910**.

(50) FIG. **13** illustrates an exploded, perspective view of a head member **202** of a printhead **104** in an illustrative embodiment. In this embodiment, head member **202** is an assembly that includes housing **230** and plate stack **232**. Plate stack **232** is affixed or attached to housing **230**, and forms one or more rows of jetting channels **402**. Housing **230** is an elongated member made from a rigid material, such as stainless steel. Housing **230** has a length (L), a width (W), and a height (H), and the dimensions of housing **230** are such that the length is greater than the width. The direction of a row of jetting channels **402** corresponds with the length of housing **230**. Housing **230** includes access hole **234** at or near its center that extends from I/O surface (not visible) through to an opposing interface surface **1312**. Access hole **234** provides passage way for an actuator assembly (not shown), such as a plurality of piezoelectric actuators, to pass through and contact diaphragms **410** of the jetting channels **402**. Interface surface **1312** is the surface of housing **230** that faces plate stack **232**, and interfaces with a plate of plate stack **232**. Housing **230** also includes manifold ducts **1316-1317** that extend longitudinally along a length of interface surface **1312**. A manifold duct **1316-1317** comprises an elongated cut or groove along interface surface **1312** that is configured to convey a print fluid, and forms at least a portion of a manifold for printhead **104**.

(51) Plate stack **232** includes a series of plates **1301-1308** that are fixed or bonded to one another to form a laminated plate structure. Plate stack **232** illustrated in FIG. **13** is intended to be an example of a basic structure of a printhead. There may be additional plates that are used in the plate stack **232** that are not shown in FIG. **13**, and the configuration of the various plates may vary as desired. Also, FIG. **13** is not drawn to scale.

(52) In this embodiment, plate stack **232** includes the following plates: a diaphragm plate **1301**, a spacer plate **1302**, a restrictor plate **1303**, chamber plates **1304-1306**, a restrictor plate **1307**, and a nozzle plate **1308**. Diaphragm plate **1301** is a thin sheet of material (e.g., metal, plastic, etc.) that is generally rectangular in shape and is substantially flat or planar. Diaphragm plate **1301** includes diaphragms **1321** comprising a sheet of a semi-flexible material that forms diaphragms **410** for the jetting channels **402**. Diaphragm plate **1301** also includes manifold openings **1322**, which are elongated apertures or holes that form part of a fluid path between a manifold and pressure chambers **412** of jetting channels **402**. Spacer plate **1302** is a thin sheet of material that is generally rectangular in shape and is substantially flat or planar. Spacer plate **1302** includes chamber openings **1324** and manifold openings **1325**. Chamber openings **1324** comprise apertures or holes that form at least part of pressure chambers **412** for jetting channels **402**. Restrictor plate **1303** is a

thin sheet of material that is generally rectangular in shape and is substantially flat or planar. Restrictor plate **1303** includes restrictor openings **1327** and manifold openings **1328**. Restrictor openings **1327** are elongated apertures or holes transversely disposed or oriented, and are configured to fluidly couple pressure chambers **412** of jetting channels **402** with a manifold. Chamber plates **1304-1306** are thin sheets of material that are generally rectangular in shape and substantially flat or planar. Chamber plate **1304** includes chamber openings **1330** and manifold openings **1331**. Chamber plate **1305** includes chamber openings **1333** and manifold openings **1334**. Chamber plate **1306** includes chamber openings **1336** and manifold openings **1337**. Restrictor plate **1307** is a thin sheet of material that is generally rectangular in shape and is substantially flat or planar. Restrictor plate **1307** includes restrictor openings **1339**, which are elongated apertures or holes transversely disposed or oriented, and are configured to fluidly couple pressure chambers **412** of jetting channels **402** with a manifold. Nozzle plate **1308** is a thin sheet of material that is generally rectangular in shape and is substantially flat or planar. Nozzle plate **1308** includes circular apertures or holes **1340** that form nozzles **414** of the jetting channels **402**. In this embodiment, nozzles **414** are arranged in two nozzle rows. However, nozzles **414** may be arranged in a single row or in more than two rows in other embodiments.

(53) FIG. **14** is a cross-sectional view of a portion of a printhead **104** in an illustrative embodiment. FIG. **14** shows a cross-section of printhead **104** along cut-plane 5-5 in FIG. **3**. Printhead **104** includes housing **230** and plate stack **232** affixed or attached to housing **230** to form jetting channels **402**. As above, plate stack **232** includes diaphragm plate **1301**, spacer plate **1302**, restrictor plate **1303**, chamber plates **1304-1306**, restrictor plate **1307**, and nozzle plate **1308**.

(54) FIGS. **15-18** disclose a printhead **104** in circulation mode in one embodiment. In circulation mode, print fluid may be re-circulated through printhead **104** past each nozzle **414**. Circulation mode may also be referred to as re-circulation mode, flow-through mode, etc. FIG. **15** is a cross-sectional view of a portion of printhead **104** in an illustrative embodiment. FIG. **15** shows a cross-section of printhead **104** along cut-plane 5-5 in FIG. **3**. In this embodiment, manifold apparatus **518** comprises manifolds **1510-1511**. Pressure chamber **412** is fluidly coupled to manifold **1510** through restrictor **520**, which controls a flow of print fluid between manifold **1510** and pressure chamber **412** along one fluid path. Pressure chamber **412** is also fluidly coupled to manifold **1511** through restrictor **522**, which controls a flow of print fluid between manifold **1511** and pressure chamber **412** along another fluid path.

(55) In this embodiment, manifold apparatus **518** further comprises a flexible separator **1520** installed, implemented, or disposed between manifolds **1510-1511**. Flexible separator **1520** comprises a membrane, wall, plate, or another structural element made from a flexible, elastic, or pliable material (e.g., plastic, rubber, thin sheet of metal, etc.) that physically separates manifold **1510** from manifold **1511**. In this embodiment, flexible separator **1520** is configured to divide manifold apparatus **518** into manifold **1510** and manifold **1511**. Manifolds **1510-1511** are fluidly isolated by flexible separator **1520** along their longitudinal lengths so that print fluid is prevented from flowing directly between manifolds **1510-1511** (although it is noted that manifolds **1510-1511** are fluidly coupled indirectly through the jetting channels **402**).

(56) The arrows in FIG. **15** illustrate a flow of print fluid through jetting channel **402**. The print fluid flows from manifold **1510** and into pressure chamber **412** through restrictor **520**. One wall of pressure chamber **412** is formed with diaphragm **410** that physically interfaces with actuator **416**, and vibrates in response to actuation by actuator **416**. The print fluid flows through pressure chamber **412** and out of nozzle **414** in the form of a droplet in response to actuation by actuator **416**. The print fluid, which is not jetted from nozzle **414**, flows from pressure chamber **412** into manifold **1511** through restrictor **522**. The print fluid that is not jetted from a nozzle **414** is referred to herein as “non-jetted print fluid”. In this scenario, manifold **1510** may be referred to as a supply manifold, as it is configured to supply print fluid to jetting channels **402**. Manifold **1511** may be referred to as a return manifold, as it is configured to receive non-jetted print fluid from jetting

channels **402**. However, the flow of print fluid may be reversed. Thus, either of manifolds **1510-1511** may act as a supply manifold or a return manifold depending the direction of flow of the print fluid. The length of restrictors **520** and **522** may be the same to allow for a reversal of flow in this manner.

(57) FIG. **16** is a schematic diagram of a printhead **104** in an illustrative embodiment. A plurality of jetting channels **402** of printhead **104** is schematically illustrated in FIG. **16** as a row of nozzles **414**. Manifold **1510** is disposed between I/O ports **211-212** that define inlets of print fluid into printhead **104**. When print fluid enters printhead **104** at one or both of I/O ports **211-212**, the print fluid flows through manifold **1510** to jetting channels **402**. Manifold **1511** is disposed between I/O ports **213-214** that define outlets of print fluid from printhead **104**. Non-jetted print fluid flows from jetting channels **402** through manifold **1511**, and exits printhead **104** at one or both of I/O ports **213-214**. Although two manifolds **1510-1511** are illustrated in FIG. **16**, a printhead **104** may include more manifolds as desired.

(58) For each of the jetting channels **402** illustrated, there is a first fluid path **601** from manifold **1510** to the jetting channel **402**, and a second fluid path **602** from the jetting channel **402** to manifold **1511**. In the embodiment shown in FIG. **15**, for example, the first fluid path **601** from manifold **1510** to jetting channel **402** may be through restrictor **520**, which controls the flow of print fluid along the first fluid path **601**. Further, the second fluid path **602** from jetting channel **402** to manifold **1511** may be through restrictor **522**, which controls the flow of print fluid along the second fluid path **602**.

(59) Flexible separator **1520** is disposed between manifold **1510** and manifold **1511**. In general, the major portions or sections of manifolds **1510-1511** are disposed longitudinally within printhead **104** to fluidly couple with the jetting channels **402**. For example, a row of jetting channels **402** is disposed longitudinally along a length of the printhead **104**. Manifolds **1510-1511** may be disposed longitudinally alongside the row of jetting channels **402**. Manifolds **1510-1511** may be horizontally aligned within printhead **104**, may be vertically aligned within printhead **104**, or may have other configurations. In this embodiment, flexible separator **1520** forms a longitudinal wall or divider between manifolds **1510-1511** so that manifolds **1510-1511** are fluidly isolated along their longitudinal lengths.

(60) FIG. **17** is a schematic diagram of manifolds **1510-1511** and a jetting channel **402** in an illustrative embodiment. FIG. **17** shows the first fluid path **601** between jetting channel **402** and manifold **1510**, and the second fluid path **602** between jetting channel **402** and manifold **1511**. In one embodiment, the length **701** of the first fluid path **601** is from an origin **711** of the pressure waves **706** to an opening **1731** of manifold **1510**. The length **702** of the second fluid path **602** is from the origin **711** of the pressure waves **706** to an opening **1732** of manifold **1511**. As above, the length **701** of the first fluid path **601** is different than the length **702** of the second fluid path **602** by a threshold length. The length **701** of the first fluid path **601** is different than the length **702** of the second fluid path **602** by the threshold length so that the arrival time of pressure waves **706** at manifolds **1510-1511** are not equal and are different by a threshold time. Also shown in FIG. **17** is flexible separator **1520** disposed between manifolds **1510-1511**. Due to the compressibility or elasticity of flexible separator **1520**, pressure waves **706** are able to communicate between manifolds **1510-1511** through flexible separator **1520**. Thus, the pressure waves **706** that arrive at manifold **1510** pass through flexible separator **1520** into manifold **1511**, and the pressure waves **706** that arrive at manifold **1511** pass through flexible separator **1520** into manifold **1510**. Because the arrival time of pressure waves **706** at manifolds **1510-1511** is different, pressure waves **706** arriving at manifold **1510** interfere destructively with pressure waves **706** arriving at manifold **1511** through flexible separator **1520**.

(61) FIG. **18** is a flow chart illustrating a method **1800** of operating printhead **104** in circulation mode in an illustrative embodiment. For method **1800**, it is assumed that printhead **104** includes a plurality of jetting channels **402** fluidly coupled to manifolds **1510-1511**, and that manifolds **1510-**

1511 are separated with a flexible separator **1520**. For each jetting channel **402**, a print fluid is conveyed from manifold **1510** to the jetting channel **402** over a first fluid path **601** (step **1802**). Non-jetted print fluid is conveyed from the jetting channel **402** to manifold **1511** over a second fluid path **602** (step **1804**). Pressure waves **706** are generated in a pressure chamber **412** of the jetting channel **402** due to actuation of an actuator **416** (step **1806**), such as to jet droplets of the print fluid from a nozzle **414** of the jetting channel **402**. The pressure waves **706** generated in the pressure chamber **412** propagate along the first fluid path **601** to manifold **1510**, and propagate along the second fluid path **602** to manifold **1511**. The design of printhead **104** produces, creates, or generates a difference in arrival time of pressure waves **706** at manifold **1510** and pressure waves **706** at manifold **1511** by a threshold time due to the difference in length **701** of the first fluid path **601** and the length **702** of the second fluid path **602** by the threshold length (step **1808**). Flexible separator **1520** provides pressure wave communication between manifolds **1510-1511** (step **1810**). Thus, the pressure waves **706** that arrive at manifold **1510** interfere destructively with the pressure waves **706** that arrive at manifold **1511** due to the communication of the pressure waves **706** through flexible separator **1520**.

(62) FIG. **19** is a schematic diagram of a printhead **104** in an illustrative embodiment. FIG. **19** is similar to FIG. **16** in that printhead **104** is schematically illustrated as including manifold **1510** disposed between I/O ports **211-212**, manifold **1511** disposed between I/O ports **213-214**, and flexible separator **1520** is disposed between manifold **1510** and manifold **1511**. Flexible separator **1520** again forms a longitudinal wall or divider between manifolds **1510-1511**. In this embodiment, flexible separator **1520** includes one or more bypass holes **1920**. A bypass hole **1920** is a hole formed through a wall or divider (e.g., flexible separator **1520**) that allows fluid to pass between manifolds **1510-1511**. Bypass holes **1920** provide a technical benefit of allowing further pressure wave communication between manifolds **1510-1511** through flexible separator **1520**.

(63) The number and placement of bypass holes **1920** shown in FIG. **19** is just an example, and may vary as desired. FIGS. **20-27** are schematic diagrams of a printhead **104** in an illustrative embodiment. FIGS. **20-23** show various examples of a printhead **104** including manifold **1510** disposed between I/O ports **211-212**, manifold **1511** disposed between I/O ports **213-214**, and flexible separator **1520** disposed between manifold **1510** and manifold **1511**. In FIG. **20**, for example, manifold **1510** has a length **2010** between a first end **2011** and a second end **2012**. When manifold **1510** has an I/O port **211-212** on ends **2011-2012** respectively, print fluid is able to flow into manifold **1510** from each end **2011-2012**. The respective flows from each end **2011-2012** intersect near the center **2014** (i.e., longitudinal center) of manifold **1510**, which creates a dead zone **2008** where there is little or no fluid flow. Similarly, manifold **1511** has a length **2020** between a first end **2023** and a second end **2024**. When manifold **1511** has an I/O port **213-214** on ends **2023-2024** respectively, print fluid is able to flow out of manifold **1511** from each end **2023-2024**. The respective flows from each end **2023-2024** create a dead zone **2018** near the center **2026** (i.e., longitudinal center) where there is little or no fluid flow. This may be an issue as the print fluid could settle or harden at dead zone **2008/2018**.

(64) In FIG. **21**, one or more bypass holes **1920** are disposed in flexible separator **1520**. For example, one or more bypass holes **1920** may be positioned at or near the longitudinal center **2140** of flexible separator **1520** (i.e., at or near the center **2014/2026** of manifolds **1510-1511**). In other words, one or more bypass holes **1920** may be disposed or positioned at or near the dead zone **2008/2018** in manifolds **1510-1511**, which creates a flow of print fluid between manifolds **1510-1511** at or near the dead zone **2008/2018**. This advantageously avoids settling or hardening of print fluid at dead zone **2008/2018**.

(65) Further, the size, placement, and/or number of bypass holes **1920** may be optimized to create or generate a uniform flow of print fluid between manifolds **1510-1511** along the length **2010/2020** of manifolds **1510-1511**. In one embodiment as shown in FIG. **22**, the size **2210** (e.g., diameter) of bypass holes **1920** may be optimized to generate a uniform flow of print fluid between manifolds

1510-1511. In one example, the size **2210** of bypass holes **1920** may be larger toward the center **2140** of flexible separator **1520**, and may decrease towards ends **2241-2242** of flexible separator **1520**. In another example, the size **2210** of bypass holes **1920** may be uniform along the length of flexible separator **1520**. In one embodiment as shown in FIG. 23, the placement of bypass holes **1920** may be optimized to generate a uniform flow of print fluid between manifolds **1510-1511**. The flow of print fluid in manifold **1510** is greater towards the ends **2011-2012** and is less toward dead zone **2008**, and the flow of print fluid in manifold **1511** is greater towards the ends **2023-2024** and is less toward dead zone **2018**. In one embodiment, a distance **2302** (i.e., longitudinal distance) between bypass holes **1920** may be shorter toward the center **2140** of flexible separator **1520**, and may increase towards ends **2241-2242** of flexible separator **1520**. In another example, the distance **2302** between bypass holes **1920** may be uniform along the length of flexible separator **1520**.

(66) FIG. 24 is a schematic diagram of a printhead **104** in an illustrative embodiment. In this embodiment, manifolds **1510-1511** are each fluidly coupled to a single I/O port. Thus, printhead **104** is schematically illustrated as including manifold **1510** fluidly coupled to I/O port **211**, manifold **1511** fluidly coupled to I/O port **212**, and flexible separator **1520** disposed between manifold **1510** and manifold **1511**. Flexible separator **1520** again forms a longitudinal wall or divider between manifolds **1510-1511**, and one or more bypass holes **1920** are formed through flexible separator **1520**.

(67) The number and placement of bypass holes **1920** shown in FIG. 24 is just an example, and may vary as desired. FIGS. 25-27 are schematic diagrams of a printhead **104** in an illustrative embodiment. FIGS. 25-27 show various examples of a printhead **104** including manifold **1510** fluidly coupled to I/O port **211**, manifold **1511** fluidly coupled to I/O port **212**, and flexible separator **1520** disposed between manifold **1510** and manifold **1511**. In FIG. 25, for example, manifold **1510** has a length **2010** between a first end **2011** and a second end **2012**. When manifold **1510** has an I/O port **211** on end **2011**, print fluid is able to flow into manifold **1510** from end **2011** and dead-ends at end **2012**. This creates a dead zone **2008** at or near end **2012** where there is little or no fluid flow. Similarly, manifold **1511** has a length **2020** between a first end **2023** and a second end **2024**. When manifold **1511** has an I/O port **212** on end **2024**, print fluid is able to flow out of manifold **1511** from end **2024**, but is dead-ended at end **2023**. This creates a dead zone **2018** at or near end **2023** where there is little or no fluid flow. This may be an issue as the print fluid could settle or harden at dead zone **2008/2018**.

(68) In FIG. 26, one or more bypass holes **1920** are disposed in flexible separator **1520**. For example, bypass holes **1920** may be disposed or positioned at or near the ends **2241-2242** of flexible separator **1520** (i.e., at or near the ends **2012/2023** of manifolds **1510-1511**). In other words, the bypass holes **1920** may be disposed at or near the dead zone **2008/2018** in manifolds **1510-1511**, which creates a flow of print fluid between manifolds **1510-1511** at or near the dead zone **2008/2018**. This advantageously avoids settling or hardening of print fluid at dead zone **2008/2018**.

(69) Further, the size, placement, and/or number of bypass holes **1920** may be optimized to create or generate a uniform flow of print fluid between manifolds **1510-1511** along the length **2010/2020** of manifolds **1510-1511**. In one embodiment as shown in FIG. 26, the size **2210** (e.g., diameter) of bypass holes **1920** may be optimized to generate a uniform flow of print fluid between manifolds **1510-1511**. In one example, the size **2210** of bypass holes **1920** may be larger toward ends **2241-2242** of flexible separator **1520**, and may decrease towards the center **2140** of flexible separator **1520**. In another example, the size **2210** of bypass holes **1920** may be uniform along the length of flexible separator **1520**. In one embodiment as shown in FIG. 27, the placement of bypass holes **1920** may be optimized to generate a uniform flow of print fluid between manifolds **1510-1511**. The flow of print fluid in manifold **1510** is greater towards end **2011** and is less toward dead zone **2008**, and the flow of print fluid in manifold **1511** is greater towards end **2024** and is less toward dead zone **2018**. In one embodiment, a distance **2302** (i.e., longitudinal distance) between bypass

holes **1920** may be shorter towards ends **2241-2242** of flexible separator **1520**, and may increase toward the center **2140** of flexible separator **1520**. In another example, the distance **2302** between bypass holes **1920** may be uniform along the length of flexible separator **1520**.

(70) In one embodiment, the flexible separator **1520** comprising bypass holes **1920** may be a filter. In this embodiment, the size **2210** of the bypass holes **1920** is small enough to capture debris that could clog nozzles **414** or narrow ink passages, and the number of bypass holes **1920** is large enough to allow print fluid to flow to pressure chambers **412** with small flow resistance. FIG. **28** is a cross-sectional view of a portion of printhead **104** in an illustrative embodiment. FIG. **28** shows a cross-section of printhead **104** along cut-plane 5-5 in FIG. **3**. As in FIG. **15**, manifold apparatus **518** is comprised of manifolds **1510-1511**. Pressure chamber **412** is fluidly coupled to manifold **1510** through restrictor **520**, which controls a flow of print fluid between manifold **1510** and pressure chamber **412** along one fluid path. Pressure chamber **412** is also fluidly coupled to manifold **1511** through restrictor **522**, which controls a flow of print fluid between manifold **1511** and pressure chamber **412** along another fluid path. In this embodiment, flexible separator **1520** comprises a filter **2820** that is installed, implemented, or disposed between manifolds **1510-1511**.

(71) FIG. **29** is a schematic diagram of a printhead **104** in an illustrative embodiment. A plurality of jetting channels **402** of printhead **104** is schematically illustrated in FIG. **29** as a row of nozzles **414**. Manifold **1510** is disposed between I/O ports **211-212** that define inlets of print fluid into printhead **104**. When print fluid enters printhead **104** at one or both of I/O ports **211-212**, the print fluid flows through manifold **1510** to jetting channels **402**. Manifold **1511** is disposed between I/O ports **213-214** that define outlets of print fluid from printhead **104**. Non-jetted print fluid flows from jetting channels **402** through manifold **1511**, and exits printhead **104** at one or both of I/O ports **213-214**. Although two manifolds **1510-1511** are illustrated in FIG. **29**, a printhead **104** may include more manifolds as desired.

(72) For each of the jetting channels **402** illustrated, there is a first fluid path **601** from manifold **1510** to the jetting channel **402**, and a second fluid path **602** from the jetting channel **402** to manifold **1511**. In the embodiment shown in FIG. **28**, for example, the first fluid path **601** from manifold **1510** to jetting channel **402** may be through restrictor **520**, which controls the flow of print fluid along the first fluid path **601**. Further, the second fluid path **602** from jetting channel **402** to manifold **1511** may be through restrictor **522**, which controls the flow of print fluid along the second fluid path **602**. Filter **2820** is disposed between manifold **1510** and manifold **1511**. Filter **2820** acts to filter the print fluid along the first fluid path **601**, and also acts as a flexible separator **1520** between manifold **1510-1511**.

(73) FIG. **30** illustrates an exploded, perspective view of a head member **202** of a printhead **104** in an illustrative embodiment. In this embodiment, housing **230** includes manifold ducts **1316-1317** along interface surface **1312**. Manifold duct **1316** comprises a groove around access hole **234** that forms part of manifold **1510** (see FIG. **15**). The major portions of manifold duct **1316** are disposed longitudinally along interface surface **1312**. Manifold duct **1317** comprises grooves toward short ends of housing **230** that form part of manifold **1511**. As before, plate stack **232** includes the following plates: diaphragm plate **1301**, spacer plate **1302**, restrictor plate **1303**, chamber plates **1304-1306**, restrictor plate **1307**, and nozzle plate **1308**. The structure of the plates may be similar to FIG. **13**. However, in this embodiment, diaphragm plate **1301** includes diaphragms **1321**, manifold openings **1322**, and flexible separator **1520**. Spacer plate **1302**, restrictor plate **1303**, and chamber plate **1304** may also include manifold openings **1322** that form part of manifold **1511**.

(74) FIG. **31** is a cross-sectional view of a portion of a printhead **104** in an illustrative embodiment. FIG. **31** shows a cross-section of printhead **104** along cut-plane 5-5 in FIG. **3**. Printhead **104** includes housing **230** and plate stack **232** affixed or attached to housing **230** to form jetting channels **402**. As above, plate stack **232** includes diaphragm plate **1301**, spacer plate **1302**, restrictor plate **1303**, chamber plates **1304-1306**, restrictor plate **1307**, and nozzle plate **1308**. In one embodiment, flexible separator **1520** in diaphragm plate **1301** physically separates manifold

1510 from manifold **1511** so that manifolds **1510-1511** are fluidly isolated by flexible separator **1520** along their longitudinal lengths and print fluid is prevented from flowing directly between manifolds **1510-1511** (although it is noted that manifolds **1510-1511** are fluidly coupled indirectly through the jetting channels **402**). In one embodiment, flexible separator **1520** includes one or more bypass holes **1920** (see FIG. **19**) that allow fluid to pass between manifolds **1510-1511**. Although shown as part of diaphragm plate **1301** in this embodiment, flexible separator **1520** is implemented in other plates in other embodiments.

(75) FIG. **32** illustrates an exploded, perspective view of a head member **202** of a printhead **104** in an illustrative embodiment. In this embodiment, housing **230** includes manifold ducts **1316-1317** along interface surface **1312**. Manifold duct **1316** comprises a groove around access hole **234** that forms part of manifold **1510** (see FIG. **15**). The major portions of manifold duct **1316** are disposed longitudinally along interface surface **1312**. Manifold duct **1317** comprises a groove around manifold duct **1316** that forms part of manifold **1511**. The major portions of manifold duct **1317** are disposed longitudinally along interface surface **1312**. As before, plate stack **232** includes the following plates: diaphragm plate **1301**, spacer plate **1302**, restrictor plate **1303**, chamber plates **1304-1306**, restrictor plate **1307**, and nozzle plate **1308**. The structure of the plates may be similar to FIG. **13**. However, in this embodiment, plate stack further includes one or more manifold plates **3209**. A manifold plate **3209** includes access hole **3234** that corresponds with access hole **234** of housing **230** to provide a passageway for electronics **204**, and manifold openings **3222**. Manifold plate **3209** also includes flexible separator **1520** between manifold openings **3222**.

(76) FIG. **33** is a cross-sectional view of a portion of a printhead **104** in an illustrative embodiment. FIG. **33** shows a cross-section of printhead **104** along cut-plane 5-5 in FIG. **3**. Printhead **104** includes housing **230** and plate stack **232** affixed or attached to housing **230** to form jetting channels **402**. As above, plate stack **232** includes manifold plate **3209**, diaphragm plate **1301**, spacer plate **1302**, restrictor plate **1303**, chamber plates **1304-1306**, restrictor plate **1307**, and nozzle plate **1308**. In this embodiment, flexible separator **1520** in manifold plate **3209** physically separates manifold **1510** from manifold **1511** so that manifolds **1510-1511** are fluidly isolated by flexible separator **1520** along their longitudinal lengths and print fluid is prevented from flowing directly between manifolds **1510-1511** (although it is noted that manifolds **1510-1511** are fluidly coupled indirectly through the jetting channels **402**). In one embodiment, flexible separator **1520** includes one or more bypass holes **1920** (see FIG. **19**) that allow fluid to pass between manifolds **1510-1511**. Although shown as part of manifold plate **3209** in this embodiment, flexible separator **1520** is implemented in other plates in other embodiments.

(77) In one embodiment, a rigid separator may be implemented, such as in manifold plate **3209**, to physically separate manifold **1510** from manifold **1511**. FIG. **34** is a cross-sectional view of a portion of a printhead **104** in an illustrative embodiment. Printhead **104** includes housing **230** and plate stack **232** affixed or attached to housing **230** to form jetting channels **402**. As above, plate stack **232** includes manifold plate **3209**, diaphragm plate **1301**, spacer plate **1302**, restrictor plate **1303**, chamber plates **1304-1306**, restrictor plate **1307**, and nozzle plate **1308**. In this embodiment, a rigid separator **3420** in manifold plate **3209** physically separates manifold **1510** from manifold **1511**. Rigid separator **3420** includes one or more bypass holes **1920** that allow fluid to pass between manifolds **1510-1511**.

(78) FIG. **35** is a schematic diagram of a design tool **3500** for a printhead **104** in an illustrative embodiment. Design tool **3500** is an apparatus or device configured to assist in the design of a printhead, such as printhead **104**. More particularly, design tool **3500** may be configured to determine one or more dimensions of components in a printhead **104**, although design tool **3500** may be configured to determine other design aspects of a printhead **104**. Design tool **3500** includes a hardware platform that includes a processor **3510** and memory **3512**. Processor **3510** comprises an integrated hardware circuit configured to execute instructions stored in memory **3512**. Memory **3512** is a non-transitory computer readable storage medium for data, instructions, etc., and is

accessible by processor **3510**. Design tool **3500** may further include a user interface **3514**. User interface **3514** is a hardware component for interacting with an end user. For example, user interface **3514** may include a display, screen, touch screen, or the like (e.g., a Liquid Crystal Display (LCD), a Light Emitting Diode (LED) display, etc.). User interface **3514** may include a keyboard or keypad, a tracking device (e.g., a trackball or trackpad), a speaker, a microphone, etc. Design tool **3500** may include various other components not specifically illustrated in FIG. **35**.

(79) FIG. **36** is a flow chart illustrating a method **3600** of designing a printhead **104** in an illustrative embodiment. The steps of method **3600** will be described with reference to design tool **3500** in FIG. **35**, but those skilled in the art will appreciate that method **3600** may be performed by other systems, tools, or entities. It is assumed for this embodiment that a printhead **104** includes or will include a manifold apparatus **518** having one or more manifolds **1510-1511**, and that manifold apparatus **518** is fluidly coupled to a plurality of jetting channels **402**. Processor **3510** plans, models, or designs a first fluid path **601** between the manifold apparatus **518** and a jetting channel **402** (step **3602**), and a second fluid path **602** between the manifold apparatus **518** and the jetting channel **402** (step **3604**). When the jetting channel **402** is in operation, pressure waves **706** are generated in a pressure chamber **412** of the jetting channel **402** due to actuation of an actuator **416**, such as to jet droplets of the print fluid from a nozzle **414** of the jetting channel **402**. The pressure waves **706** generated in the pressure chamber **412** will propagate along the first fluid path **601** to manifold apparatus **518**, and propagate along the second fluid path **602** to manifold apparatus **518**. Processor **3510** selects, calculates, or identifies a target difference in arrival time of the pressure waves **706** at manifold apparatus **518** (step **3606**). Processor **3510** selects a difference in length between the first fluid path **601** and the second fluid path **602** by a threshold length that causes the target difference in arrival time of the pressure waves **706** at manifold apparatus **518** (step **3608**). The difference in length **701** of the first fluid path **601** and length **702** of the second fluid path **602** by the threshold length will cause a difference in arrival time of pressure waves **706** at manifold apparatus **518** (i.e., by a threshold time). Processor **3510** may then configure the first fluid path **601** and the second fluid path **602** for the jetting channels **402** based on the threshold length (step **3610**). In one embodiment, processor **3510** may display or otherwise provide the threshold length (optional step **3620**) to a user through user interface **3514**, over a network to a remote system, or perform other functions when selecting the target length. In one embodiment, processor **3510** may control, regulate, set, or instruct one or more fabrication processes to fabricate the printhead **104** based on the threshold length between the fluid paths **601-602** (optional step **3622**).

(80) In one embodiment, processor **3510** may determine the resonant frequency or Helmholtz frequency of the jetting channels **402** (optional step **3616**), and select the target difference in arrival time of the pressure waves **706** at manifold apparatus **518** based on the resonant frequency (optional step **3618**). For example, processor **3510** may perform a test on printhead **104** or a similar printhead (i.e., another printhead with jetting channels having the same or similar dimensions), or may receive test data regarding the printhead **104** or a similar printhead to determine the resonant frequency of the jetting channels **402**. Processor **3510** may perform a simulation on printhead **104** or a similar printhead, or may receive simulation data regarding the printhead **104** or a similar printhead to determine the resonant frequency of the jetting channels **402**. Processor **3510** may determine the resonant frequency of jetting channels **402** in other ways. Processor **3510** may then select the target difference in arrival time and/or threshold length based on the resonant frequency of the jetting channels **402**. For example, the target difference in arrival time may be a half resonant cycle (e.g., 0.5) or half Helmholtz cycle of the jetting channels **402**, or a multiple of the half resonant cycle (e.g., 1.5, 2.5, 3.5, etc.).

(81) Embodiments disclosed herein can take the form of software, hardware, firmware, or various combinations thereof. In one particular embodiment, software is used to direct a processing system of design tool **3500** to perform the various operations disclosed herein. FIG. **37** illustrates a processing system **3700** operable to execute a computer readable medium embodying programmed

instructions to perform desired functions in an illustrative embodiment. Processing system **3700** is operable to perform the above operations by executing programmed instructions tangibly embodied on computer readable storage medium **3712**. In this regard, embodiments can take the form of a computer program accessible via computer-readable medium **3712** providing program code for use by a computer or any other instruction execution system. For the purposes of this description, computer readable storage medium **3712** can be anything that can contain or store the program for use by the computer.

(82) Computer readable storage medium **3712** can be an electronic, magnetic, optical, electromagnetic, infrared, or semiconductor device. Examples of computer readable storage medium **3712** include a solid-state memory, a magnetic tape, a removable computer diskette, a random access memory (RAM), a read-only memory (ROM), a rigid magnetic disk, and an optical disk. Current examples of optical disks include compact disk-read only memory (CD-ROM), compact disk-read/write (CD-R/W), and DVD.

(83) Processing system **3700**, being suitable for storing and/or executing the program code, includes at least one processor **3702** coupled to program and data memory **3704** through a system bus **3750**. Program and data memory **3704** can include local memory employed during actual execution of the program code, bulk storage, and cache memories that provide temporary storage of at least some program code and/or data in order to reduce the number of times the code and/or data are retrieved from bulk storage during execution.

(84) Input/output or I/O devices **3706** (including but not limited to keyboards, displays, pointing devices, etc.) can be coupled either directly or through intervening I/O controllers. Network adapter interfaces **3708** may also be integrated with the system to enable processing system **3700** to become coupled to other data processing systems or storage devices through intervening private or public networks. Modems, cable modems, IBM Channel attachments, SCSI, Fibre Channel, and Ethernet cards are just a few of the currently available types of network or host interface adapters. Display device interface **3710** may be integrated with the system to interface to one or more display devices, such as printing systems and screens for presentation of data generated by processor **3702**.

(85) Although specific embodiments were described herein, the scope of the invention is not limited to those specific embodiments. The scope of the invention is defined by the following claims and any equivalents thereof.

Claims

1. A printhead comprising: a manifold apparatus comprising a first manifold, a second manifold, and a flexible separator disposed between the first manifold and the second manifold; and a plurality of jetting channels each fluidly coupled to the first manifold via a first fluid path and to the second manifold via a second fluid path; wherein when pressure waves generated in pressure chambers of the jetting channels propagate along the first fluid path and the second fluid path, the pressure waves communicate between the first manifold and the second manifold through the flexible separator; wherein a design of the first fluid path and the second fluid path causes a difference in arrival time of the pressure waves at the manifold apparatus to produce destructive interference of the pressure waves at the manifold apparatus.
2. The printhead of claim 1 wherein: the design comprises lengths of the first fluid path and the second fluid path differing by a threshold length so that the arrival time of the pressure waves at the manifold apparatus are different by a threshold time.
3. The printhead of claim 2 wherein: the threshold time is based on a resonant frequency of the jetting channels.
4. The printhead of claim 3 wherein: the threshold time is approximately a half resonant cycle or a multiple of the half resonant cycle.
5. The printhead of claim 1 wherein: the design comprises material properties of the printhead that

form the first fluid path and the second fluid path differing so that the arrival time of the pressure waves at the manifold apparatus are different.

6. The printhead of claim 1 wherein: the design comprises a volume of the first fluid path and the second fluid path differing so that the arrival time of the pressure waves at the manifold apparatus are different.

7. The printhead of claim 1 wherein: the flexible separator fluidly isolates the first manifold and the second manifold.

8. The printhead of claim 1, wherein: the flexible separator comprises one or more bypass holes that fluidly couple the first manifold and the second manifold.

9. The printhead of claim 8 wherein: a size of the bypass holes is larger toward a longitudinal center of the flexible separator, and decreases towards ends of the flexible separator.

10. The printhead of claim 8 wherein: a distance between the bypass holes is shorter toward a longitudinal center of the flexible separator, and increases towards ends of the flexible separator.

11. The printhead of claim 8 further comprising: a single Inlet/Outlet (I/O) port fluidly coupled to the first manifold; and a single I/O port fluidly coupled to the second manifold; wherein the bypass holes are disposed near ends of the flexible separator.

12. The printhead of claim 8 further comprising: a single Inlet/Outlet (I/O) port fluidly coupled to the first manifold; and a single I/O port fluidly coupled to the second manifold; wherein a size of the bypass holes is larger toward ends of the flexible separator, and decreases towards a longitudinal center of the flexible separator.

13. The printhead of claim 8 further comprising: a single Inlet/Outlet (I/O) port fluidly coupled to the first manifold; and a single I/O port fluidly coupled to the second manifold; wherein a distance between the bypass holes is shorter toward ends of the flexible separator, and increases towards a longitudinal center of the flexible separator.

14. A jetting apparatus comprising: at least one printhead of claim 1.

15. A method of operating a printhead comprising a plurality of jetting channels configured to jet a print fluid, the method comprising: for each jetting channel of the plurality, conveying the print fluid from a first manifold of a manifold apparatus to the jetting channel over a first fluid path; and conveying non-jetted print fluid from the jetting channel to a second manifold of the manifold apparatus over a second fluid path; generating pressure waves in a pressure chamber of the jetting channel that propagate along the first fluid path and the second fluid path; and communicating the pressure waves between the first manifold and the second manifold through a flexible separator disposed between the first manifold and the second manifold; wherein a design of the first fluid path and the second fluid path causes a difference in arrival time of the pressure waves at the manifold apparatus to produce destructive interference of the pressure waves at the manifold apparatus.

16. The method of claim 15 further comprising: the design producing a difference in the arrival time of the pressure waves at the manifold apparatus by a threshold time due to a difference in length between the first fluid path and the second fluid path by a threshold length.

17. The method of claim 16 wherein: the threshold time is based on a resonant frequency of the jetting channels.

18. The method of claim 17 wherein: the threshold time is approximately a half resonant cycle or a multiple of the half resonant cycle.

19. A printhead comprising: a manifold apparatus comprising a first manifold, a second manifold, and a separator that separates the first manifold and the second manifold, wherein the separator is flexible to communicate pressure waves between the first manifold and the second manifold; and a plurality of jetting channels each fluidly coupled to the first manifold via a first fluid path and to the second manifold via a second fluid path; wherein when the pressure waves generated in pressure chambers of the jetting channels propagate along the first fluid path and the second fluid path, features of the first fluid path and the second fluid path cause a difference in arrival time of

- the pressure waves at the manifold apparatus to produce destructive interference of the pressure waves at the manifold apparatus.
20. The printhead of claim 19 wherein: lengths of the first fluid path and the second fluid path are different by a threshold length so that the arrival time of the pressure waves at the manifold apparatus are different by a threshold time.
21. A jetting apparatus comprising: at least one printhead of claim 19.
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